Supporting Information

Can Structure-Based Models Clear Untested PFAS? External Validation Reveals Endpoint-, Representation-, and Distribution-Dependent Reliability

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1 Reproducible data lock

All counts in the revised manuscript derive from the files under `canonical_results/` in the public repository (`https://github.com/bisu9082/pfas-clearance-validation`). Canonical isomeric SMILES were generated with RDKit. PFAS were removed from broad training by CAS match, PFASSTRUCTV6 canonical-structure match, or direct measured-PFAS structure match. Identical desalted structures in the broad data were represented once, using the minimum log toxicity across duplicate registry rows. The inventory was independently deduplicated and measured PFAS were removed before it was described as "untested."

Table 1: Data-lock reconciliation.

Quantity	Count
EnviroTox effect records	80,912
EnviroTox substance entries	4,267
Parsable broad chemical records	4,068
Duplicate canonical-structure rows among parsable records	213
Broad rows matching PFAS	73
Unique non-PFAS training structures	3,788
High-hazard training structures ($\leq$ 0.1 mg/L)	936
Measured PFAS structures	138
High-hazard measured PFAS	57
Raw PFASSTRUCTV6 inventory records	19,028
Inventory records with invalid SMILES	22
Inventory records overlapping measured PFAS	140
Unique parsable untested inventory structures	18,851
Untested structures within the applicability domain	17,128

The 140 overlapping inventory rows are not expected to equal the 138 measured PFAS one-to-one because inventory records can match a measured substance through alternative registry or structural forms. The exact matching rows are retained in `inventory_measured_overlap.csv`.

2 Model and conformal settings

The default model used 2,048-bit ECFP4 fingerprints and a random forest with 300 trees, `max_features=sqrt`, and seed 20260620. Broad cross-validation used 160 trees per fold, five

folds, and five repeats. Conformal validation used 40 repeats. Broad marginal calibration used a random 80/20 non-PFAS train/calibration split and evaluated the fixed PFAS external set. PFAS-conditional validation used a fixed model trained on all non-PFAS structures and disjoint 69/69 PFAS calibration/test halves. Repeated-split percentile ranges are reported because test observations recur across repeats.

Table 2: Repeated-split coverage and interval width on PFAS. Values are means across 40 repeats; ranges in parentheses are the 2.5th–97.5th percentiles across repeats. Width is in $\log_{10}$ mg/L.

Nominal	Marginal coverage	Conditional coverage	Marginal width	Conditional width
0.50	0.214 (0.188–0.247)	0.526 (0.362–0.668)	1.94	3.83
0.60	0.283 (0.239–0.326)	0.630 (0.463–0.755)	2.41	4.62
0.70	0.385 (0.319–0.457)	0.724 (0.593–0.843)	2.97	5.88
0.80	0.530 (0.471–0.580)	0.828 (0.753–0.942)	3.73	7.09
0.90	0.659 (0.623–0.703)	0.922 (0.841–0.986)	4.84	10.53
0.95	0.740 (0.696–0.805)	0.972 (0.899–1.000)	5.83	14.46

At nominal 0.90, no known-low-hazard PFAS in the held-out conditional test halves was cleared in any repeat. This is an out-of-split power check: each tested substance was not used in the calibration half for that repeat.

3 External endpoint and representation sensitivity

Table 3: External PFAS performance after structure locking.

Endpoint	Representation	Training n	PFAS n	R^2	Spearman ρ
Most-sensitive	ECFP4 binary	3,788	138	−0.68	−0.03
Most-sensitive	Morgan counts	3,788	138	−0.62	−0.02
Most-sensitive	RDKit descriptors (165)	3,788	138	−0.06	+0.42
Acute-only	ECFP4 binary	3,438	97	−0.06	+0.30
Acute-only	Morgan counts	3,438	97	−0.02	+0.32
Acute-only	RDKit descriptors (165)	3,438	97	+0.23	+0.48
Chronic-only	ECFP4 binary	1,714	108	−0.68	+0.02

The paired endpoint comparison used the same 97 PFAS. The most-sensitive rank correlation was +0.047 and the acute-only correlation was +0.296, giving $\Delta\rho = +0.250$ with substance-level bootstrap 95% CI $[+0.090, +0.418]$. The per-configuration sensitivity fits in this section (e.g., most-sensitive ECFP4, $R^2 = -0.68$, $\rho = -0.03$) use the repeated cross-validation protocol and differ slightly from the single full-training deployment fit reported in the main text ($R^2 = -0.58$, $\rho = -0.07$); both are negative and rank-indistinguishable from zero.

4 Applicability domain and deployment

The training leave-self-out five-neighbor threshold was 5.887 in Euclidean binary-ECFP4 space. Of 18,851 unique untested structures, 17,128 were in domain. Cross-fitted broad marginal calibration

produced median in-domain 90% width 4.09 log units and classified 3,138 structures as clearable. Because this calibration achieved only 0.659 mean PFAS coverage at nominal 0.90, that count is reported as an invalid-on-PFAS comparison, not a decision output. Calibration on all 138 measured PFAS produced median in-domain width 10.12 log units and zero clearable structures.

Deployment conditional coverage requires exchangeability between measured and untested PFAS. The present data cannot prove this assumption. Therefore, the inventory intervals are an assumption-qualified extrapolation, while the repeated-split coverage results are the empirical validation claim.

5 Algorithm, applicability-domain, and calibration robustness

The external negative transfer on the protective (most-sensitive) endpoint is not specific to random forests. Re-fitting the same structure-locked non-PFAS training set with two alternative learners on identical ECFP4 features reproduced the failure (Table 4).

Table 4: Algorithm sensitivity of external PFAS validation (most-sensitive endpoint, binary ECFP4, $n_{\text{train}} = 3{,}788$, $n_{\text{PFAS}} = 138$). All learners fail externally.

Learner	R^2	Spearman ρ
Random forest (canonical)	−0.580	−0.071
Gradient boosting	−0.596	−0.041
Ridge regression	−0.280	+0.086

The applicability-domain membership of the untested inventory is moderately sensitive to the distance threshold, but the deployment conclusion is not: no in-domain structure was clearable at any threshold (Table 5).

Table 5: Applicability-domain sensitivity. In-domain counts among the 18,851 unique untested structures at three percentiles of the training leave-self-out five-neighbour distance. The manuscript uses the 95th percentile.

Training-distance percentile	Threshold	In domain	In domain (%)
90th	5.253	15,037	79.8
95th (used)	5.887	17,128	90.9
99th	6.872	18,295	97.1

Finally, the zero-clearance result is stable to the calibration sample: under leave-one-PFAS-out recalibration of the conditional 90% quantile, the quantile varied with a coefficient of variation of 0.8%, and the in-domain clearable count remained zero in all 138 leave-one-out recalibrations. The deployment conclusion is therefore not an artifact of a single calibration draw.

6 Files supporting the numerical audit

- `canonical_metrics.json`: final manuscript metrics.
- `conformal_repeated_split.csv`: all repeat-level coverage and width results.
- `conformal_summary.csv`: the values in Table S2.

- `endpoint_representation_results.csv`: Table S3 values.
- `inventory_measured_overlap.csv`: records removed from the untested inventory.
- `untested_pfas_marginal_and_conditional90.csv`: structure-level deployment output.